

GARDEN PROCUREMENT, BUDGETING & SCHOOL PACKAGE SYSTEM

[Starter](#) | [Standard](#) | [Advanced](#) | [Research](#)

Purpose	Turn the 50 GrowMeal garden designs into practical procurement and deployment packages.
Workbook	A companion cost model provides editable quantities, unit costs, contingency, installation and annual operating budget calculations.
Pricing	No market prices are hard-coded as official quotations. Schools/LIFEWS enter current supplier prices.
Scope	Infrastructure, irrigation, tools, inputs, learning resources, training, installation, logistics, maintenance and replenishment.

FOUR SCHOOL PACKAGE TIERS

Package	Purpose	Typical scope	Core inclusion	Cost
STARTER	Entry implementation	1-2 priority gardens	Basic hand tools; watering; planting materials; labels; selected student/teacher resources	Low
STANDARD	Full level implementation	Up to 10 gardens for one educational level	Garden infrastructure; irrigation basics; tool/storage set; measurement kit; complete level resources; teacher orientation	Moderate
ADVANCED	Multi-level implementation	20-30 gardens across multiple levels	Expanded irrigation; stronger storage; measurement/data equipment; soilless/vertical options; staff training; M&E; tools	Moderate-High
RESEARCH	Advanced investigation platform	Advanced JSS/SSS gardens and selected integrated systems	Instrumented gardens; sensors; data logging; controlled experiments; advanced soilless/irrigation; research consumables; specialist training	High

1. PURPOSE & PROCUREMENT PRINCIPLES

Purpose	Convert GrowMeal technical designs into budgetable, quotable and deployable school packages.
Procurement rule	Buy only infrastructure that the school can safely operate, maintain, water and use instructionally.
Local sourcing	Prioritize safe, durable and repairable local materials where they meet performance requirements.
Cost control	Separate one-time capital costs from recurring consumables, maintenance and training costs.
Quotation rule	Actual quotations should use current supplier prices, transport, taxes, installation labor and site-specific conditions.

2. COST ARCHITECTURE

CAPEX	Site preparation, beds/structures, tanks, irrigation, durable tools, storage, measurement equipment, sensors and signage.
Initial consumables	Growing medium, compost, seeds/planting materials, labels, small fittings and hygiene supplies.
Implementation services	Site survey, fabrication, transport, installation, commissioning and initial teacher orientation.
OPEX	Replacement planting materials, irrigation repairs, consumables, maintenance, printing/replacement learning materials and refresher support.
Contingency	Use a clearly stated contingency percentage rather than hiding uncertainty inside unit prices.

3. PACKAGE SIZING

Small school	Use a Starter or compact Standard package and prioritize the gardens most useful for the participating classes.
Medium school	Use a Standard or Advanced package with dedicated storage and a more reliable water/irrigation arrangement.
Large school	Use multiple level packages or an Advanced package with centralized storage, maintenance responsibility and shared infrastructure.
Research/demonstration site	Use the Research package only where staff, data management, maintenance and technical supervision can support advanced systems.

4. PROCUREMENT WORKFLOW

Step 1	Confirm school level(s), learner numbers, available footprint, water source and implementation objective.
Step 2	Complete the site survey and select the garden/package configuration.
Step 3	Generate a preliminary BOM and obtain supplier quotations.
Step 4	Compare quotations on specification, durability, delivery, warranty/after-sales support and total cost - not price alone.
Step 5	Approve budget and procurement schedule.
Step 6	Receive and inspect materials before installation.
Step 7	Install, commission and record actual quantities/costs.
Step 8	Transfer durable assets to the school inventory and establish the maintenance/replacement budget.

5. SUPPLIER EVALUATION

Specification compliance	Does the item meet the required dimensions, material, capacity and safety requirement?
Durability	Can it withstand the expected school environment and repeated learner use?
Repairability	Can local staff obtain fittings/spares and repair it economically?
Delivery	Can the supplier meet the required deployment date and location?
Support	Is installation, warranty, technical support or replacement available where needed?
Value	Compare total usable life and serviceability rather than lowest purchase price alone.

6. SCHOOL QUOTATION STRUCTURE

Section A	Garden infrastructure and fabrication.
Section B	Water and irrigation.
Section C	Tools, storage and safety.
Section D	Growing inputs and planting materials.
Section E	Measurement, sensors and technical equipment.
Section F	Learning resources and teacher materials.
Section G	Training, installation and commissioning.
Section H	Transport/logistics, contingency and applicable taxes/fees.

7. INSTALLATION LABOR & LOGISTICS

Labor	Estimate site preparation, fabrication, assembly, plumbing/irrigation, finishing, signage and commissioning separately.
Transport	Budget vehicle type, distance, loading/offloading and protection of fragile equipment.
Staging	Deliver materials in installation sequence where storage is limited.
Acceptance	Record shortages, damage and substitutions before accepting final delivery.
Site variation	Any change from the approved specification should be documented with cost and technical implications.

8. ANNUAL OPERATING BUDGET

Planting cycle	Seeds/seedlings and crop-specific growing inputs.
Water system	Emitters, connectors, hose, seals, valves and minor tank/plumbing repairs.
Tools	Replacement of damaged/lost low-cost tools and sharpening/repair where appropriate.
Learning	Replacement labels, printing, journals/cards where consumable, measurement consumables and approved food-literacy materials.
Maintenance	Structural repairs, paint/protection where appropriate, growing-medium replenishment and pest-management materials permitted by school policy.
Training	New-teacher induction and targeted refresher training.

9. INVENTORY & ASSET CONTROL

Asset tagging	Assign an asset or inventory identifier to durable equipment where practical.
Custody	Record responsible school unit/person for storage and issue/return.
Condition	Record good, repair, damaged, lost or decommissioned status.
Consumables	Track separately from durable assets to support realistic annual replenishment.
End of term	Reconcile inventory before ordering replacements.

10. PACKAGE CHANGE CONTROL

Substitution	Do not substitute materials solely to reduce price if safety, food-contact suitability, stability or learning function is reduced.
Scope change	Record added/removed gardens and update the BOM, installation plan and teacher-resource requirement.
Price change	Keep unit prices date-stamped because supplier markets change.
Approval	Material scope or cost changes should be approved according to LIFEWS/school procurement authority.

11. DEPLOYMENT PHASING

Phase 1	Readiness, site survey, package selection and budget approval.
Phase 2	Procurement and fabrication.
Phase 3	Delivery and installation.
Phase 4	Commissioning and teacher orientation.
Phase 5	12-week implementation.
Phase 6	Term review, maintenance and replenishment.
Scale-up	Add new levels only after the current package is operational and maintainable.

12. FINANCIAL CONTROL

Budget baseline	Freeze an approved baseline before procurement starts.
Commitments	Track purchase orders/approved commitments separately from actual payments.
Variance	Compare budget, committed and actual cost by category.
Documentation	Retain quotations, approvals, receipts/invoices, delivery notes and acceptance records.
Closeout	Reconcile final cost, outstanding liabilities, unused materials and assets transferred to the school.

APPENDIX A - QUOTATION APPROVAL CHECK

School	_____
Package / levels	_____
Site survey complete	Yes / No
BOM version/date	_____
Supplier quotations compared	_____
CAPEX subtotal	_____
Implementation/logistics	_____
Contingency	_____
Total approved budget	_____
Annual OPEX estimate	_____
Approval / date	_____

APPENDIX B - PROCUREMENT CLOSEOUT

Approved budget	
Actual final cost	
Variance	
Gardens installed	
Assets transferred	
Outstanding defects	
Unused materials	
Maintenance budget established	Yes / No
Closeout approved	